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Industrial Internet of Things (IIoT) Training

Elevate your team's Industrial IoT (IIoT) expertise with our dynamic, hands-on course designed to master the latest IoT and cloud technologies. Build over 10 innovative projects using ESP32, Arduino, ThingSpeak, Blynk, AWS Cloud, Node-Red, Raspberry Pi, Python, InfluxDB, Grafana, and various sensors. This course seamlessly integrates theory with practical experience, empowering your organization to harness the full potential of IIoT and ignite innovation. Enroll today to revolutionize your industrial IoT capabilities.

Syllabus

Module-1: Introduction to Internet of Things

- Introducing to IOT
- IOT Applications
- IOT Network Architecture
- IOT Device Architecture
- IOT Communication Protocols
- IOT Product Development Overview

Hands-on Sessions

- Installation of Software, Tools and Account Creations

Module-2: Getting Started with Arduino

- A tour of Arduino Board and Hardware: Power Supply, Power Pins,
- Analog and Digital Pins
- Types of Arduino Boards
- Introduction to Arduino programming
- Variables,IF-Else conditional statements
- Loops: For, While
- Functions, Digital Inputs and Digital Outputs
- The serial monitor, Arrays and strings
- Using Libraries in Arduino

Hands-on Sessions:

- Interfacing LED, Blinking LED Project
- Traffic Lights Control Project
- Interfacing Buzzer and producing different sounds

Module-3: Sensors, Actuators & Electronics

- Introduction to sensors and types
- Analog Sensors: Temperature, Light Sensor, Potentiometer,
- Digital Sensors: Door sensor, Motion Sensor, DHT11 sensor and Push Button
- Digital Signals
- Basic electronics: resistors, capacitors, diodes, transistors and etc.,
- Introduction to Actuators, Working with various actuators

Hands-on Sessions:

- Interfacing LDR and reading Light Sensor Data
- Interfacing thermistor and reading temperature values
- Interfacing DHT11 sensor and reading temperature, humidity values
- Reading Push button status
- Working with Accelorometer
- Automatic street Lighting system

Module-4: Wireless Communication Technologies

- Introduction to wireless technologies: WiFi, Bluetooth, Ethernet, LoRaWAN, WiMAX and ZigBee

Hands-on Sessions:

- Scanning nearby WiFi access points by using ESP32
- Connecting to the WiFi Access Point and Internet Access

Module-5 : IoT using Blynk Mobile Platform

- Setting up Blynk
- Install Blynk Library
- Exploring various control widgets
- Exploring various display widgets
- Notification widgets and virtual pins

Hands-on Sessions:

- Smart Home Project using Blynk
- Smart Environment Monitoring using Blynk

Module-6 : Programming ESP32 using Micropython

- Introducing MicroPython
- MicroPython Hardware
- How to Program in MicroPython
- MicroPython Libraries

Hands-on Sessions:

- MicroPython programming examples
- Blinking an Led project, Traffic Lights Control
- Working with DHT11 sensor

Module-7 : Smart Applets suing IFTTT

- Introduction, Automate day to day activities through IFTTT
- Posting updates on Facebook
- Automation with IFTTT, Sending text message notifications
- Google Voice to control Home Appliances

Hands-on Sessions:

- Sending weather data to email
- Voice Controlled Home Automation using Google Assistant
- Smart Light Bulb Design using RGB Led
- Notification Cube Project

Module-8: Cloud Data Monitoring using Arduino

- Concept & Architecture of Cloud
- Tools, API and Platform for integration of IoT devices with Cloud
- Internet of Things platforms for Arduino

Hands-on Sessions:

- Creating Project in Thingspeak platform
- Weather station using Thingspeak Cloud

Module-9: IoT using AWS

- Getting Started with AWS IOT Platform
- Send Sensor data to AWS IoT using MQTT
- Reading sensor data in ASWS IoT Core
- Getting started with Node-RED
- Reading data in Node-Red
- Create Node-RED application to receive events from the device

Hands-on Sessions:

- Smart Home Project using AWS and Node-Red
- Notifications using AWS SNS Service

Module 10: Introduction to Industrial IoT

- Introduction to Industrial IOT
- Significance of IIoT in Modern Industries
- Key Components and Technologies in IIoT

Module 11: Getting Started with Raspberry Pi

- Introduction to Raspberry Pi Board
- Installing Raspberry Pi OS
- Raspberry Pi Configuration
- Networking in Raspberry Pi
- Introduction to Raspberry Pi OS
- Linux Commands
- Linux File System
- Virtual Network Computing

Module 12: Python Programming on Raspberry Pi

- Introduction to Python Programming
- Python Variables, Input/Output Functions & Operators
- Conditional Statements and Loops
- Strings, Lists, Dictionaries, Tuples Python Examples

Module 13: Interfacing Hardware & Cloud with Raspberry Pi

- RPi.GPIO Module & Blinking Led Program
- Traffic Lights Control System
- Introduction to Sensors and Actuators
- Working with Buzzer
- Working with DHT11 Sensor
- Working with Actuators
- Sending Temperature and Humidity Values to Thingspeak Cloud
- Sending Alerts or Notifications

Module 14: Communication Protocols for Industrial Devices

- Overview of industrial communication protocols (Modbus, RS232, OPC UA, Profibus, CAN)
- Implementing protocols for data exchange and control in industrial environments
- Ensuring compatibility and integration with Raspberry Pi

Module 15: RS232 UART (Universal Asynchronous Receiver-Transmitter)

- Introduction to RS232 UART
- UART Tools and Utilities
- Interfacing NEO-6M GPS Module with Raspberry Pi

Module 16: RS485 Modbus

- Introduction to Modbus
- Modbus Tools and Utilities
- Integrating Temperature and humidity sensor with Modbus Transmitter (XY-MD02) on Raspberry Pi

Module-17: OPC Unified Architecture (OPC UA)

- Introduction to OPC UA
- Implementing OPC UA Communication between Windows and Raspberry Pi
- Real-world Applications and Use Cases

Module-18: MQTT (Message Queuing Telemetry Transport)

- MQTT protocol
- MQTT clients: MQTT Fx, MyMQTT for Android
- Download Eclipse Mosquitto Broker in windows
- Mosquito Broker on Raspberry Pi
- Establishing an MQTT connection between a Windows and Raspberry Pi.
- Raspberry Pi to MQTTfx and MyMQTT
- Establishing connection between MQTT fx and AWS IoT Core
- MQTT protocol between Thingspeak – ESP 32 interfaced with DHT 11

Module-19: CoAP (Constrained Application Protocol)

- Introduction to CoAP
- Thingsboard cloud environment
- Establishing CoAP connection between Raspberry Pi and CoAP supported cloud and monitoring DHT sensor data

Module-20: InfluxDB Essentials

- What is InfluxDB ?
- InfluxDB Basics
- Creating an InfluxDB Account
- Real-world Applications and Use Cases

Module-21: Data Collection and Management with InfluxDB

- Introduction to InfluxDB & Installation
- Sensor Data Collection
- Data Retrieval and Queries
- Optimizing data storage in InfluxDB.
- Handling large volumes of time-series data efficiently.
- Implementing Server Less Approach with IoT using InfluxDB.

Module-22: Real-Time Data Processing

- Python Integration with InfluxDB
- Real-Time Analysis Techniques
- Utilizing Matplotlib and Plotly for real-time visualization.
- Creating dynamic visualizations from live data streams.
- Optimizing Python code for real-time processing.

Module-23 Visualization with Grafana

- Grafana Setup and Installation
- Designing interactive dashboards in Grafana.
- Adding panels to visualize IoT data.
- Alerting and Monitoring

Module-24: IoT for Smart Factories: Building a Future-Ready Industrial Solution

- Learn how IoT transforms Smart Factories.
- IoT for Smart Factories project concept

Module-25: Vibration Monitoring with Raspberry Pi and MPU6050

- Understanding MPU6050
- Interface MPU6050 sensor with Raspberry Pi
- Sending data to Influxdb
- Visualizing sensor data using Grafana

Module 26: Energy Monitoring and BLE Communication

- "Introduction to BLE Communication
- Interfacing PZEM04T Energy Sensor
- Design and implement an ESP32-based BLE energy meter
- Energy Meter Data logging into InfluxDB
- Grafana Data Visualisation"

Final Project: Machine Monitoring System

Tools & Technologies

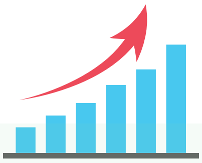
Training Highlights

- **Course Completion Certificate**
- **60 Hours Live sessions**
- **10+ Course Projects**
- **20+ Practical assignments**
- **Capstone Projects**

IoT Development Kit (for training)

- NodeMCU (ESP32)
- Breadboard
- Relay Module
- DHT11 Sensor
- USB Cable
- Jumper Wires
- Light Sensor
- RGB LED
- Push Button
- Buzzer
- LED's, Resistors
- MPU6050 Accelerometer

Why enroll for IoT Course?



By 2020, 50 billion devices will be connected via the Internet of Things (IoT)

Average salary of IoT Developer is $150k - Glassdoor.com

The global IoT market is expected to grow from $655.8 billion to $1.7 trillion in 2020

Additional Benefits to students

CISCO — Internet of Things Certification

udemy — IoT using Raspberry Pi Certification

Actual Course Fee:

RS 20000/- per participant

RS 10000/- per participant

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Capstone Project (Hands-on session)

Participant will work on Capstone Project as a team. List of some projects:

1. Smart Shoe Project using ESP32
2. Smart Street Lighting System using Blynk
3. Design and Implementation of Fall Detection System Using MPU6050
4. Smart Dustbin using ESP32
5. Machine Health Monitoring System
6. Smart Irrigation Control Project using ESP32, AWS and Node-Red
7. Baby Monitoring system using MPU6050, Temperature, wetness sensor
8. Machine Monitoring System
9. Air Quality Monitoring System
10. Indoor Gardening using IoT
11. Door security system for home monitoring based on ESp32
12. ESP32 Based Electric Energy Consumption Monitoring system
13. IoT based industrial wireless fans
14. Mushroom Farm Automation Using Internet of Things (IoT)
15. IoT based Smart bottle for healthcare
16. Smart Drip Irrigation Control System using IoT
17. Smart Wireless Display using IoT (OLED/LCD Display)
18. Smart Kitchen using IoT
19. Wireless Blood Oxygen (SPO2) Monitoring System
20. Intelligent Home Project

and more....